

ECI 016 - Final Project

High Risk Roads in Sacramento

Report

By: Nathan Ward

Executive Summary: This project aims to identify the most at-risk streets in Sacramento, excluding freeways and highways. Where risk is defined as the probability of a vehicle collision. High-risk streets were identified by analyzing Sacramento 2023 collision data. Where all highways and freeways were filtered out, and a list of all streets with 30+ collisions was recorded in a year. A pre-screening filter to focus on roads with enough data for statistical significance. From this point, AADT (Annual Average Daily Traffic) data were used to quantify the volume of each street experiences, and street lengths were queried in R from OSM (OpenStreetMap). Risk is then quantified as $R = \frac{n}{AADT * 365 * L} * 10^7$

Where n = number of crashes, AADT = annual average daily traffic, L = length of street, and 10^7 is a decimal point adjustment parameter. These calculated crash rate values were then compared to the critical crash rate equation,

which is defined as $R_c = R_a + K \left(\frac{R_a}{M} \right)^{\frac{1}{2}} + \frac{1}{2M}$, where R_a = average crash rate, K = probability factor, and M = exposure = $\frac{AADT * 365 * L}{10^7}$. Where an R_a value of 0.9 and K value of 1.645 (95% confidence) were chosen for this project as industry standards. Streets with an R value greater than R_c are then defined as high-risk roads, which should be examined to improve safety conditions on these roads Figure 5. shows a table of reduced results. Further analysis was then conducted to determine the percentage of crashes with respect to time of day. If a correlation between peak hour and an increase in crash percentage is found, it can be hypothesized that crash risk will decrease if road congestion/volume-to-capacity ratios are reduced. Defining peak hours from 4-6 PM, Figure 4. shows that approximately 22% of collisions happen within this timeframe, which is only 13% of the day. My findings suggest that roads such as Folsom Blvd, Stockton Blvd, and Florin Rd are high-priority zones in need of engineering intervention with respect to safety due to their high crash density.

Introduction

Background: Traffic safety is an integral part of Sacramento's urban planning, particularly with respect to high-volume areas and within the framework of the City's Vision Zero action plan [5]. Which aims to mitigate traffic fatalities and serious injuries, with a primary focus on high-volume roadways which serve as a backbone to any city's transportation network.

Motivation: While total collision count provides insight into the magnitude of collisions per road, identifying specific roads that are at-risk with high crash rate frequencies through statistical rate analysis allows for precision in infrastructure investment. By distinguishing between random collision occurrences and fundamental systematic roadway design flaws, traffic engineers can prioritize high-impact safety measures.

Significance: Pinpointing areas of high-risk regarding vehicle collisions provides empirical evidence to justify CIP (Capital Improvement Projects). This data-driven strategy is essential for Sacramento to meet its Vision Zero safety goals and reduce economic impact due to collisions.

Statement of Problem / Hypothesis

Problem: Raw collision counts are insufficient for identifying roadway safety hazards, as this methodology does not account for traffic volume or exposure. Without spatial analysis and rate-based normalization, it is difficult to distinguish between random, isolated collisions and systematic collisions due to faults in road design that lead to high-risk corridors.

Hypothesis: Spatial data analysis on collisions in each area will reveal a disproportionate number of crashes that are concentrated on a small subset of roads. Specifically, these corridors will exhibit an actual crash rate greater than the critical crash rate with a 95% confidence level. Particularly during the defined peak hours of 4-6 PM.

Data Sources:

1. [2023 California Collision Data](#): The primary data source, which contained 1 year's worth of collision data in Sacramento, with key variables being primary street name, time of day, day of week, and longitudinal/latitudinal data. About 8,000 collision records were recorded in Sacramento, approximately 1000 of which were due to these high-risk roads.
2. [2023 California AADT Data](#): Contained AADT data for each roadway being analyzed, which is used to find the volume of traffic flowing through each road.
3. [Sacramento Centerline Data](#): Contained longitudinal and latitudinal data of each roadway being analyzed, which was used to map collisions and high-risk roadways.
4. [Open Street Map](#): Contained length data of roadways being analyzed.

Spatial Analysis and Visualization

Assumptions:

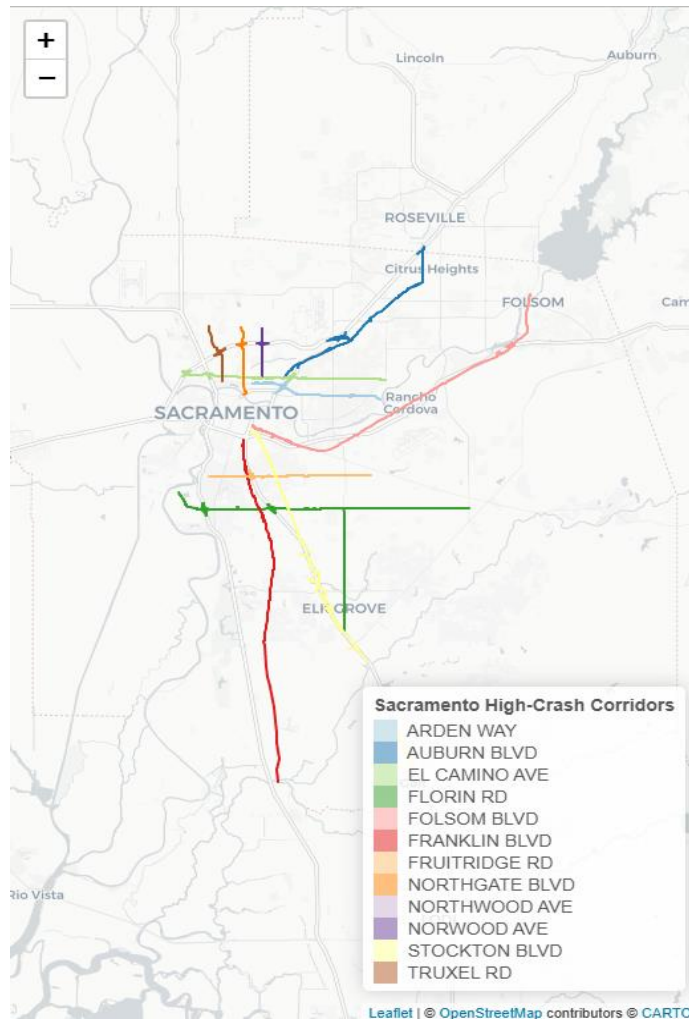
1. AADT data is constant throughout 2023
2. Back and ahead AADT values are equal (They are not equal but were averaged as such, as collision data did not state if the vehicle was traveling ahead or behind)
3. AADT data can be seen as approximately constant from the average of all strips of road.
4. GPS coordinates in TMS dataset are accurate within 50ft.
5. Sacramento peak hour is 4-6 PM

Spatial Analysis / Data Reduction:

1. An Excel spreadsheet was created that contained all road names in collision data, and each was classified by highway/freeway or road type. Both were imported into R, and all highways/freeways were removed from the collision data.
2. Used R to sum collisions by road name into a new table named road_crash_summary.
3. Imported AADT data and averaged AADT data by road name, then averaged ahead and behind AADT.
4. Matched road names from AADT data and road names of road_crash_summary with collisions greater than or equal to 30. (15 values obtained)
5. Used R and OSM to input the length of road data for potential high-risk roads.
6. Calculated risk factor with $R = \frac{n}{AADT * 365 * L} * 10^7$, Where n = number of crashes, AADT = annual average daily traffic, L = length of street, and 10^7 is a decimal point adjustment parameter.
7. Calculated the critical crash rate with $R_c = R_a + K \left(\frac{R_a}{M} \right)^{\frac{1}{2}} + \frac{1}{2M}$, where R_a = average crash rate, K = probability factor, and M = exposure = $\frac{AADT * 365 * L}{10^7}$. Where an R_a value of 0.9 and K value of 1.645 (Industry standards).
8. Defined at-risk roads as roads where $R > R_a$. (12 Values)
9. Used R to make graphs of collisions with respect to time of day and day of week to find trends in collisions with respect to peak hours.

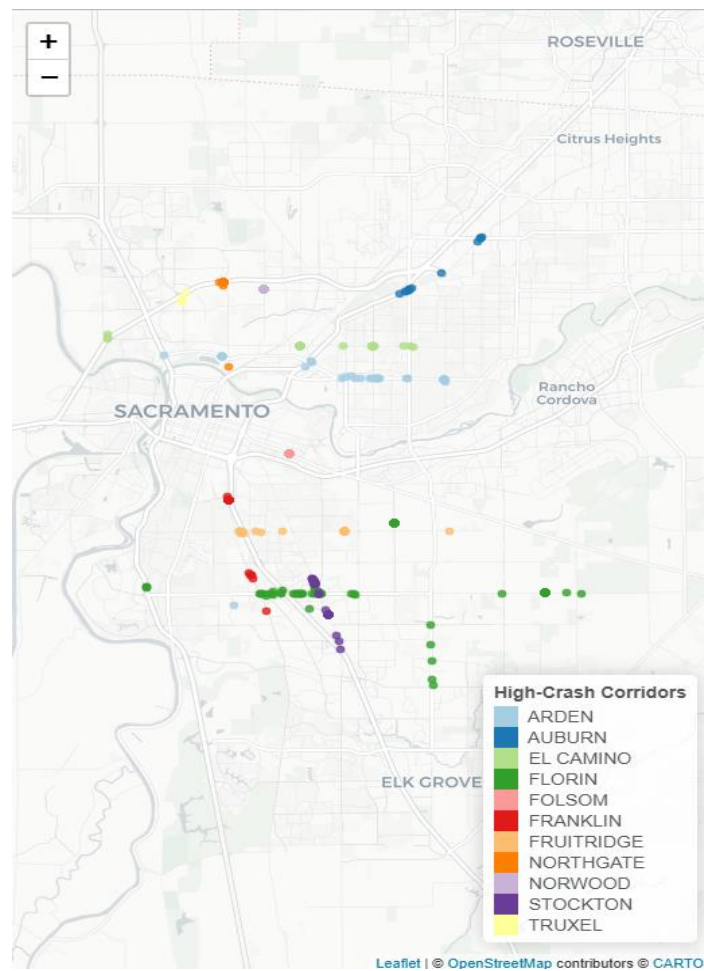
Visualization:

Figure 1



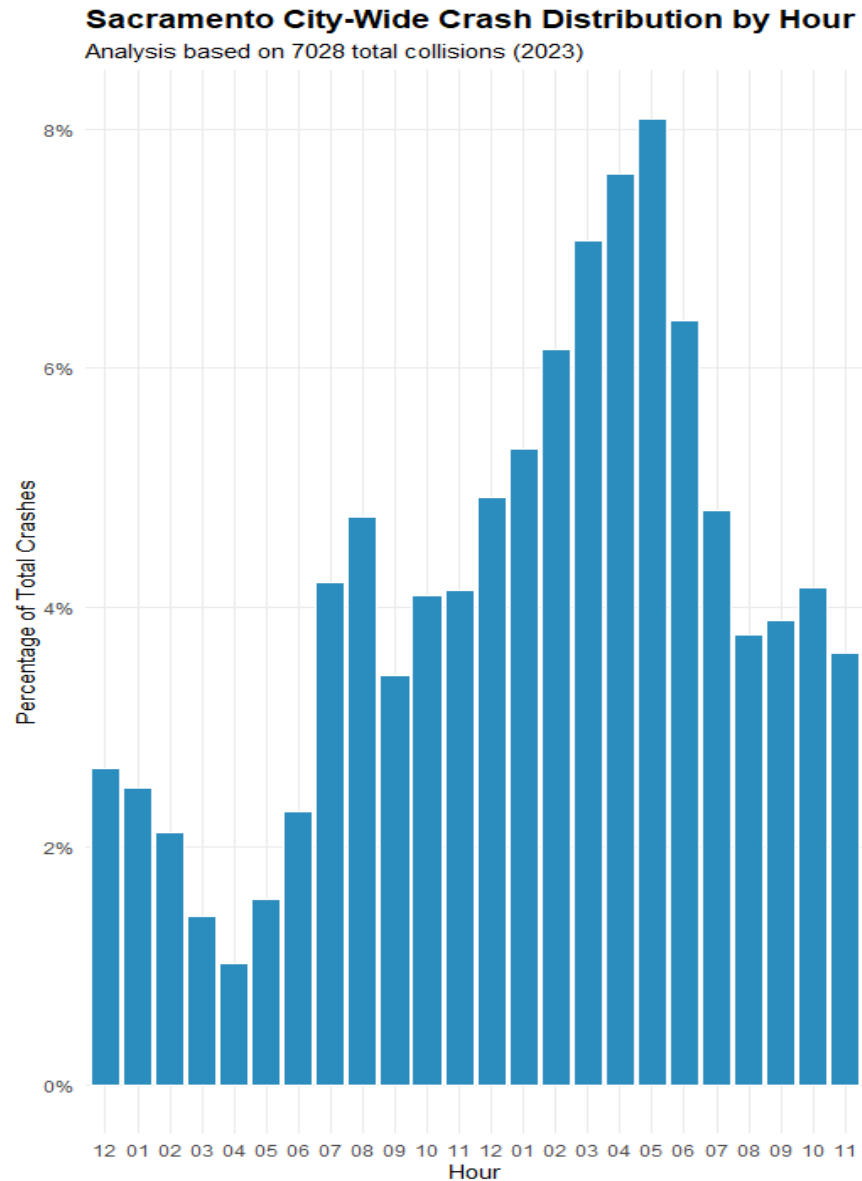
**Figure 1 shows a map of all the high-risk roads, each highlighted in its own color. The map shows a sort of radial connectivity, with high-risk streets branching out from central Sacramento.*

Figure 2



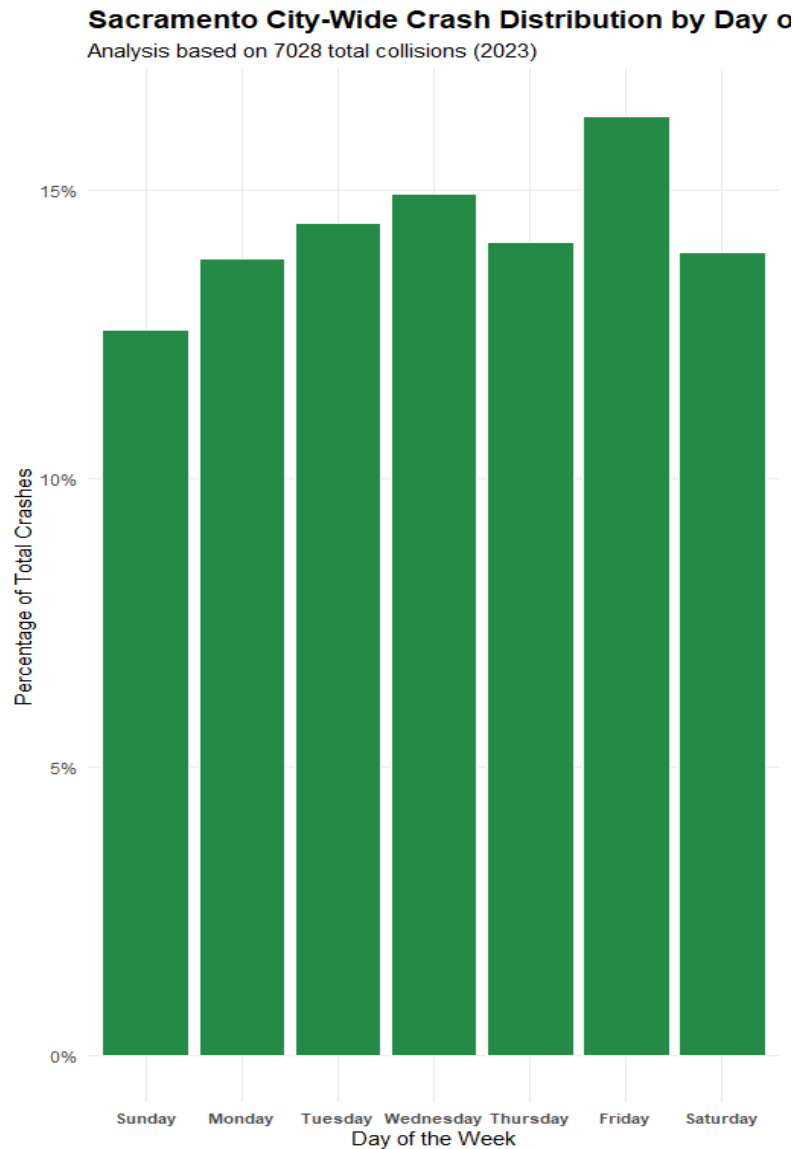
**Figure 2 shows a map of all crashes in each high-risk road plotted.*

Figure 3



**Figure 3 shows a bar graph of normalized collisions in terms of percentages, showing a large semi-linear rise between 11 and 6, and approximately 22% of crashes occur during peak hours (4-6 PM). Which indicates that an increase in volume and traffic congestion leads to a higher collision rate.*

Figure 4.



**Figure 4 shows a bar graph of normalized collisions in terms of percentage with respect to the day of the week. The high extrema is Friday, consisting of about 16.5% of crashes, and the low extrema is Monday, consisting of about 12.5% of crashes, with normalized values at approximately 14.3%.*

Figure 5.

Rd	ADDT	Crashes	Length	Crashes/ 10 Million VMT	Critical Rate	C-Cr	Final Decision
NORWOOD AVENUE	182000	66	2.95	3.367888	1.333299	2.034589	REQUIRES INTERVENTION
FOLSOM BOULEVARD	99000	78	7.4	2.916989	1.27499	1.641999	REQUIRES INTERVENTION
NORTHGATE BOULEVARD	147000	64	4.59	2.598706	1.289459	1.309247	REQUIRES INTERVENTION
TRUXEL ROAD	144000	80	6.04	2.519983	1.246855	1.273128	REQUIRES INTERVENTION
FRANKLIN BOULEVARD	64500	85	14.44	2.50034	1.236355	1.263985	REQUIRES INTERVENTION
FLORIN ROAD	92500	165	20.98	2.329399	1.145248	1.184151	REQUIRES INTERVENTION
STOCKTON BOULEVARD	174250	130	9.17	2.228991	1.165968	1.063024	REQUIRES INTERVENTION
FRUITRIDGE ROAD	162500	127	11.07	1.934238	1.15308	0.781158	REQUIRES INTERVENTION
AUBURN BOULEVARD	129000	51	5.27	2.055308	1.288107	0.767201	REQUIRES INTERVENTION
EL CAMINO AVENUE	181000	84	7.06	1.800956	1.192634	0.608322	REQUIRES INTERVENTION
ARDEN WAY	164000	92	8.96	1.715312	1.175591	0.539721	REQUIRES INTERVENTION
HOWE AVENUE	139000	46	9.81	0.924232	1.184551	-0.26032	OKAY
WATT AVENUE	148633	85	18.6	0.84236	1.112627	-0.27027	OKAY
FULTON AVENUE	139500	34	7.03	0.949853	1.228699	-0.27885	OKAY

**Figure 5 shows a table of all the reduced data, where the final decision is deemed to require engineering safety intervention if the $C > Cr$ (the crash rate is greater than the critical crash rate).*

Discussion and Conclusion: The analysis confirms that a small group of roads carries the highest risk in Sacramento city. The temporal data shows a significant increase in collisions during peak hours. The high density of collisions on these roads suggests that reducing conflict points at intersections, managing speeds, and adding lanes through design could lower collision frequency by allowing for a higher flux of vehicles to travel per unit time while reducing traffic congestion.

Contributions:

Nathan Ward	Found data and references Wrote R scripts to reduce data Wrote R scripts for temporal distributions Wrote R scripts for geographical distribution Wrote report
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References:

1. [2023 California Collision Data](#)
2. [2023 California AADT Data](#)
3. [Sacramento Centerline Data](#)
4. [Open Street Map:](#)
5. [Vision Zero](#)

Appendices:

A1. Reduces data to a number of collisions per road, instead of each collision having its own row.

```
# Create A summary of number of crashes per road
library(tidyverse)

road_crash_summary <- Collision_Data_Roads_Reduced %>%
  group_by(PrimaryRoad) %>%
  summarise(Total_Crashes = n()) %>%
  arrange(desc(Total_Crashes))

print(head(road_crash_summary, 20))

View(road_crash_summary)

library(tidyverse)

street_dictionary <- c(
  "\\bBLVD\\b" = "BOULEVARD",
  "\\bRD\\b"   = "ROAD",
  "\\bAVE\\b"  = "AVENUE",
  "\\bST\\b"   = "STREET",
  "\\bDR\\b"   = "DRIVE",
  "\\bWY\\b"   = "WAY",
  "\\bCIR\\b"  = "CIRCLE",
  "\\bPKWY\\b" = "PARKWAY",
  "\\bHWY\\b"  = "HIGHWAY",
  "\\bCT\\b"   = "COURT"
)

road_crash_summary <- road_crash_summary %>%
  mutate(
    Expanded_Road_Name = str_replace_all(PrimaryRoad, street_dictionary)
  )
!
print(head(road_crash_summary))
```

A2. Adds road length data to high-risk roads, and adds averaged AADT data.

```
#Script to fetch road length data from road_crash_summary
library(tidyverse)
library(osmdata)
library(sf)

my_top_roads <- head(road_crash_summary$Expanded_Road_Name, 54)

get_road_length <- function(road_name) {
  cat("Fetching map data for:", road_name, "...\\n")

  search_name <- str_to_title(road_name)
```

```
query <- opq(bbox = "Sacramento, California") %>%
  add_osm_feature(key = "name", value = search_name) %>%
  add_osm_feature(key = "highway")

osm_data <- tryCatch({
  osmdata_sf(query)
}, error = function(e) {
  return(NULL)
})

Sys.sleep(2)

if (is.null(osm_data) || is.null(osm_data$osm_lines)) {
  return(tibble(Expanded_Road_Name = road_name, Length_Miles = NA))
}

total_miles <- sum(as.numeric(st_length(osm_data$osm_lines)) / 1609.34,
na.rm = TRUE)

return(tibble(Expanded_Road_Name = road_name, Length_Miles =
round(total_miles, 2)))
}

lengths_table <- map_df(my_top_roads, get_road_length)

road_crash_summary <- road_crash_summary %>%
  left_join(lengths_table, by = "Expanded_Road_Name")

print(head(road_crash_summary, 10))

#Adds road name column
library(tidyverse)

ADDT <- ADDT %>%
  mutate(
    road_name = str_split_i(DESCRIPTION, ",", 1)
  )

View(ADDT)

#Creates averaged AADT Data
library(tidyverse)

ADDT_cleaned <- ADDT %>%
  mutate(across(
    c(BACK_PEAK_HOUR, BACK_PEAK_MADT, BACK_AADT,
      AHEAD_PEAK_HOUR, AHEAD_PEAK_MADT, AHEAD_AADT),
    ~as.numeric(gsub("[^0-9.]", "", .))
  ))

ADDT_averaged <- ADDT_cleaned %>%
  group_by(road_name) %>%
```

```
summarise(
  Avg_Back_Peak_Hour = mean(BACK_PEAK_HOUR, na.rm = TRUE),
  Avg_Back_Peak_MADT = mean(BACK_PEAK_MADT, na.rm = TRUE),
  Avg_Back_AADT      = mean(BACK_AADT, na.rm = TRUE),

  Avg_Ahead_Peak_Hour = mean(AHEAD_PEAK_HOUR, na.rm = TRUE),
  Avg_Ahead_Peak_MADT = mean(AHEAD_PEAK_MADT, na.rm = TRUE),
  Avg_Ahead_AADT      = mean(AHEAD_AADT, na.rm = TRUE),

  .groups = "drop"
) %>%
mutate(across(where(is.numeric), ~round(., 0)))

View(ADDT_averaged)
```

A3. Filters collisions by road type (road/freeway)

```
#filter by road type
install.packages("writexl")
library(tidyverse)
library(writexl)

road_type_clean <- Road_Type %>%
  distinct(`Rd/Frw`, .keep_all = TRUE)

roads_only_df <- Collision_Data %>%
  left_join(road_type_clean, by = c("PrimaryRoad" = "Rd/Frw")) %>%

  filter(Type == "Road")

print(table(roads_only_df$Type))

#Average AADT Data and add to crash summary
library(tidyverse)

ADDT_final <- ADDT_averaged %>%
  mutate(
    Total_Average_AADT = round((Avg_Back_AADT + Avg_Ahead_AADT) / 2, 0)
  ) %>%
  select(road_name, Total_Average_AADT)

road_crash_summary <- road_crash_summary %>%
  left_join(ADDT_final, by = c("Expanded_Road_Name" = "road_name"))

View(road_crash_summary)
```

A4. Creates maps of collisions, high-risk roadways, and bar graphs of collisions per hour, collisions per day of week.

```
# Map Crashes
library(tidyverse)
library(leaflet)

target_roads <- c(
  "NORTHWOOD", "NORWOOD", "FOLSOM", "EL CAMINO", "NORTHGATE",
  "TRUXEL", "FRANKLIN", "STOCKTON", "AUBURN", "FRUITRIDGE",
  "ARDEN", "FLORIN"
)

road_pattern <- paste(target_roads, collapse = "|")

map_data <- Collision_Data %>%
  mutate(
    PrimaryRoad = str_trim(toupper(PrimaryRoad)),
    SecondaryRoad = str_trim(toupper(SecondaryRoad))
  ) %>%
  filter(str_detect(PrimaryRoad, road_pattern)) %>%
  mutate(MainRoad = str_extract(PrimaryRoad, road_pattern)) %>%
  filter(!is.na(Latitude) & !is.na(Longitude))

pal <- colorFactor(
  palette = "Paired",
  domain = map_data$MainRoad
)

leaflet(map_data) %>%
  addProviderTiles(providers$CartoDB.Positron) %>%
  addCircleMarkers(
    lng = ~Longitude,
    lat = ~Latitude,
    radius = 3,
    color = ~pal(MainRoad), # Use the palette for the circle border
    fillColor = ~pal(MainRoad), # Use the palette for the fill
    stroke = TRUE,
    weight = 1,
    fillOpacity = 0.8,
    popup = ~paste0(
      "<b>Road:</b> ", MainRoad, "<br>",
      "<b>Intersection:</b> ", PrimaryRoad, " & ", SecondaryRoad, "<br>",
      "<b>Type:</b> ", `Collision Type Description`
    )
  ) %>%
  addLegend(
    position = "bottomright",
    pal = pal,
    values = ~MainRoad,
    title = "High-Crash Corridors",
    opacity = 1
  )

# Map High Risk Streets
target_roads <- c(
  "ARDEN WAY", "AUBURN BLVD", "EL CAMINO AVE", "FLORIN RD", "FOLSOM BLVD",
```

```
"FRANKLIN BLVD", "FRUITRIDGE RD", "NORTHGATE BLVD", "NORTHWOOD AVE",  
"NORWOOD AVE", "STOCKTON BLVD", "TRUXEL RD"  
)  
  
road_pattern <- paste0("\\b", target_roads, "\\b", collapse = "|")  
  
highlighted_roads <- Sacramento_Geometry %>%  
  mutate(FULLSTREET = toupper(FULLSTREET)) %>%  
  
  filter(  
    str_detect(FULLSTREET, road_pattern) &  
    !str_detect(FULLSTREET, "ALLEY")  
  ) %>%  
  
  mutate(MainRoad = str_extract(FULLSTREET, road_pattern)) %>%  
  
  filter(!is.na(MainRoad))  
pal <- colorFactor(palette = "Paired", domain = target_roads)  
  
leaflet(highlighted_roads) %>%  
  addProviderTiles(providers$CartoDB.Positron) %>%  
  addPolylines(  
    color = ~pal(MainRoad),  
    weight = 2,  
    opacity = 1.0,  
    popup = ~FULLSTREET,  
    label = ~MainRoad  
  ) %>%  
  addLegend(  
    position = "bottomright",  
    pal = pal,  
    values = target_roads,  
    title = "Sacramento High-Crash Corridors"  
  )  
  
  
# Crashes By Hour  
library(tidyverse)  
library(lubridate)  
library(scales)  
  
hourly_all_crashes <- Collision_Data %>%  
  filter(!is.na(`Crash Date Time`)) %>%  
  
  mutate(  
    DateTime = as.POSIXct(`Crash Date Time`, format="%Y-%m-%d %H:%M:%S"),  
    Hour24 = hour(DateTime),  
  
    HourLabel = format(strptime(Hour24, format="%H"), "%I")  
  ) %>%  
  
  mutate(HourFactor = factor(Hour24, levels = 0:23)) %>%  
  
  group_by(HourFactor, HourLabel) %>%  
  summarise(Count = n(), .groups = "drop") %>%
```

```
mutate(Percentage = (Count / sum(Count)))

total_crashes <- sum(hourly_all_crashes$Count)
cat("Total Sacramento Crashes Analyzed:", total_crashes, "\n")

ggplot(hourly_all_crashes, aes(x = HourFactor, y = Percentage)) +
  geom_col(fill = "#2b8cbe", color = "white") +
  # Map the 24 distinct hours to the 12-hour labels
  scale_x_discrete(labels = hourly_all_crashes$HourLabel) +
  scale_y_continuous(labels = percent_format(accuracy = 1)) +
  theme_minimal() +
  labs(
    title = "Sacramento City-Wide Crash Distribution by Hour",
    subtitle = paste0("Analysis based on ", total_crashes, " total
collisions (2023)"),
    x = "Hour",
    y = "Percentage of Total Crashes"
  ) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(size = 9)
  )

# Crashes by Day of Week
library(tidyverse)
library(scales)

daily_all_crashes <- Collision_Data %>%
  filter(!is.na(`Day Of Week`)) %>%

  mutate(`Day Of Week` = factor(`Day Of Week`,
                                levels = c("Sunday", "Monday", "Tuesday",
"Wednesday",
                                "Thursday", "Friday",
"Saturday")) %>%
  group_by(`Day Of Week`) %>%
  summarise(Count = n()) %>%
  mutate(Percentage = (Count / sum(Count)))

total_crashes <- sum(daily_all_crashes$Count)
cat("Total Sacramento Crashes Analyzed:", total_crashes, "\n")

ggplot(daily_all_crashes, aes(x = `Day Of Week`, y = Percentage)) +
  geom_col(fill = "#238b45", color = "white") +
  scale_y_continuous(labels = percent_format(accuracy = 1)) +
  theme_minimal() +
  labs(
    title = "Sacramento City-Wide Crash Distribution by Day of Week",
    subtitle = paste0("Analysis based on ", total_crashes, " total
collisions (2023)"),
    x = "Day of the Week",
    y = "Percentage of Total Crashes"
```

```
) +  
theme(  
  plot.title = element_text(face = "bold", size = 14),  
  panel.grid.minor = element_blank(),  
  axis.text.x = element_text(size = 8, face = "bold")  
)
```

End of Report.